RAG-Based Medical Knowledge Assistant

NLP & Retrieval-Augmented Generation Project using Merck Manuals

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Executive Summary

- Built a low-code AI medical assistant using Merck Manuals
- Compared LLM-only vs Prompt Engineering vs RAG
- RAG improved groundedness for most queries
- System demonstrates potential clinical and operational value
- Validated using LLM-as-judge: groundedness + relevance scoring.

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Business Problem Overview

What's broken today:

- Healthcare professionals face information overload
- Critical decisions must be made under time pressure
- Medical knowledge is fragmented across long manuals
- Risk of inconsistent care and delayed diagnosis

Why this matters:

- Delays and errors directly affect patient outcomes
- Clinicians need answers with traceable source support
- Standardized, evidence-grounded guidance can reduce variability in care decisions.

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Solution Approach

High-level approach:

- Use Large Language Models (LLMs) for natural-language QA
- Enhance reliability using Retrieval-Augmented Generation (RAG)
- Ground answers in Merck Medical Manuals
- RAG = retrieve relevant manual passages + generate an answer grounded in them.
- RAG adds traceability + reduces hallucinations by grounding answers in an approved medical source.

Architecture:

- PDF → Chunking → Embeddings → Vector DB → Retriever → LLM → Answer
- Development progression: LLM-only → Prompt Engineering → RAG

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Question Answering using LLM

Model Used:

- TheBloke/Mistral-7B-Instruct-v0.2-GGUF | mistral-7b-instruct-v0.2.Q6_K.gguf
- Loaded from Hugging Face (GGUF via llama-cpp).

Why Selected:

- Instruction-tuned for structured Q&A
- Efficient 7B parameter model
- Compatible with RAG architecture
- Suitable for controlled inference environments

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Question Answering using LLM

1. Sepsis Management
   - Early recognition
   - IV antibiotics
   - Fluid resuscitation
   - Vasopressors if needed
   - Monitor lactate/urine output/MAP and organ function trends.

2. Appendicitis
   - Symptoms: RLQ pain, fever, nausea
   - Standard treatment: Appendectomy
   - Antibiotics may be considered in selected uncomplicated cases (clinical judgment required).

3. Patchy Hair Loss
   - Causes: Alopecia areata, fungal infection, thyroid disorders or autoimmune
   - Treatments: Steroids, antifungals, immunotherapy

4. Traumatic Brain Injury
   - Stabilize ABCs
   - CT imaging
   - Control intracranial pressure
   - Rehabilitation therapy

5. Leg Fracture
   - Immobilization
   - Avoid weight bearing
   - Casting or surgery
   - Physical therapy for recovery

Comments / Observations
- Responses were medically structured and coherent
- Answers were general but not sourced from Merck Manual
- No traceability or citation support
- Potential risk of hallucination in complex protocols
- Produces clinically plausible output, but cannot guarantee guideline-level correctness without grounding.

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Question Answering using LLM with Prompt Engineering

System Prompt: You are a clinical medical assistant. Provide structured, step-by-step, medically accurate responses. Avoid speculation. Clearly distinguish symptoms, diagnosis, and treatment. If uncertain, state limitations explicitly.

Parameter Tuning: Parameters Tested:

- Temperature, Max Tokens, Top-p, Top-k

Best Performing Parameters Run#3: Selected based on structured output, clinical completeness, and appropriate uncertainty handling.

- Temperature: 0.2
- Max Tokens: 350
- Top-p: 0.9
- Top-k: 40

Why These Parameters Worked:

- Lower temperature → More deterministic and clinically consistent responses
- Higher max tokens → Complete, well-structured explanations
- Controlled sampling → Reduced variability and hallucination

Experiments conducted using Sepsis query as representative clinical scenario

Run	Temperature	MaxTokens	Top-p	Top-k	Outcome
1	0.0	250	0.9	40	Balanced length; Stepwise; More assertive
2	0.3	350	0.95	50	Balanced length; Stepwise; Cautious
3	0.2	350	0.9	40	Balanced length; Stepwise; Cautious
4	0.6	300	0.95	50	Balanced length; Stepwise; More assertive
5	0.2	200	0.85	30	Balanced length; Stepwise; More assertive

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Question Answering using LLM with Prompt Engineering

Prompt-Engineered Answers: LLM-only with improved system prompt — no document retrieval

1. Sepsis Management
   - Early recognition using clinical signs and SOFA/qSOFA criteria
   - Obtain blood cultures before antibiotics
   - Administer broad-spectrum IV antibiotics promptly
   - Initiate aggressive IV fluid resuscitation
   - Use vasopressors if hypotension persists
   - Monitor lactate levels and organ function

2. Appendicitis
   - Symptoms: right lower quadrant pain, nausea, vomiting, fever
   - Diagnosis via clinical exam and imaging
   - Standard treatment: appendectomy
   - Antibiotics may be used in selected uncomplicated cases

3. Patchy Hair Loss
   - Common causes: alopecia areata, fungal infection, thyroid disorders
   - Treatment depends on cause: corticosteroids, antifungals, or immunotherapy
   - Evaluation may include clinical exam and laboratory testing

4. Traumatic Brain Injury
   - Immediate ABC stabilization
   - Neurological assessment and CT imaging
   - Manage intracranial pressure if elevated
   - Rehabilitation for functional recovery

5. Leg Fracture
   - Immobilize and avoid weight bearing
   - Control bleeding if open fracture
   - Imaging confirmation
   - Casting or surgical fixation
   - Physical therapy during recovery

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Prompt Engineering — Observations & Limitations

Observations

- Responses were more structured and clinically organized
- Improved prioritization of diagnostic and treatment steps
- Reduced vague or overly generalized language
- More consistent formatting across all 5 queries
- Improved clarity and step-by-step reasoning
- Still not grounded in external medical sources (Merck Manual content not used)
- No citation traceability → residual hallucination risk remains
- Need external grounding (Merck) to improve traceability + reduce hallucinations.

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Data Preparation for RAG

Dataset Used: medical_diagnosis_manual.pdf

- Based on the Merck Medical Manual
- ~4,000+ pages of structured medical knowledge

Purpose:

- Provide authoritative, domain-specific grounding for responses.

Text Splitting Parameters:

Splitter: RecursiveCharacterTextSplitter (tiktoken encoder) - to prepare the document for retrieval

- Chunk Size: 1000 characters
- Chunk Overlap: 200 characters
- Total Chunks Created: 4685

Why:

- Ensures context completeness
- Preserves medical explanations across chunk boundaries
- Improves retrieval relevance

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Data Preparation for RAG

Embedding Model Used:

- SentenceTransformer – all-MiniLM-L6-v2

Purpose:

- Efficient and lightweight
- Strong semantic similarity performance
- Suitable for large document indexing

Vector Database

- Chroma DB
- Persisted locally as medical_db

Purpose: Store embeddings for similarity-based retrieval.

RAG Configuration Parameters

- Retriever k: 3
- Max Tokens: 350
- Temperature: 0.2
- Retrieval method: similarity search
- Embedding dimension: 384 (MiniLM-L6-v2)

Purpose:

- k=3 balances relevance and noise
- Low temperature maintains deterministic, clinical tone
- Adequate token length ensures complete medical responses

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Question Answering using RAG

System Prompt (RAG Version)
"You are a clinical medical assistant. Answer the question using only the provided context from the Merck Manual. If the information is not available in the context, clearly state that it is not found."

User Prompt Template
Context: {retrieved_chunks}
Question: {user_query}
Provide a structured, medically accurate answer based strictly on the context.

Retrieval Configuration

- Text Splitter: RecursiveCharacterTextSplitter
- Chunk Size: 1000
- Chunk Overlap: 200
- Total Chunks Created: 4685
- Retriever Type: Similarity Search
- k (Top Results Retrieved): 3

Why:

- Balanced context size with overlap for medical continuity
- k=3 minimizes noise while preserving relevance

Generation Parameters

- Model: Mistral-7B-Instruct-v0.2 (GGUF)
- Temperature: 0.2
- Max Tokens: 350
- Top-p: 0.9
- Top-k: 40

Why:

- Low temperature ensures clinical consistency
- Token length sufficient for structured medical responses

Tuning Experiments (Sepsis Query Representative Test Case)

Run	k	Temperature	Max Tokens	Observation
1	1	0.2	300	Insufficient contextual grounding
2	3	0.2	350	Best balance of completeness & focus
3	5	0.2	350	Excess contextual overlap; increased verbosity
4	3	0.4	350	Slight variability in phrasing
5	3	0.2	250	Response length constrained; incomplete protocol steps

Final configuration selected: Run 2 (k=3, temp=0.2, max_tokens=350) based on contextual completeness and reduced redundancy.

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Question Answering using RAG

RAG Answers (Merck-Manual Grounded)
Answers generated using retrieved Merck chunks
RAG Settings Used: chunk_size=1000 | overlap=200 | k=3 | temp=0.2 | max_tokens=350

1) Sepsis Management

- Identify sepsis early and assess organ dysfunction
- Start broad-spectrum antibiotics promptly (after cultures if feasible)
- Give IV fluids for hypotension / poor perfusion
- If shock persists, initiate vasopressors and monitor lactate/organ function
- Provide supportive ICU care (oxygen/ventilation/renal support as needed)

2) Appendicitis — Symptoms & Treatment

- Typical symptoms: abdominal pain (often migrating to RLQ), nausea/vomiting, fever, tenderness
- Diagnosis supported by clinical exam + imaging
- Definitive treatment: appendectomy
- Antibiotics may be used in selected uncomplicated cases, but surgery remains standard

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Question Answering using RAG (continued)

3) Sudden Patchy Hair Loss — Causes & Treatment

- Common causes include alopecia areata and fungal infection (tinea capitis)
- Evaluation focuses on identifying etiology (history/exam ± labs)
- Treatment depends on cause: corticosteroids for alopecia areata; antifungals if infection suspected
- Address contributing factors (autoimmune/thyroid considerations)

4) Traumatic Brain Injury (TBI)

- Immediate ABC stabilization and neurological assessment
- Imaging (CT) to identify bleeding/swelling
- Manage intracranial pressure when indicated
- Supportive care + structured rehabilitation (physical/cognitive) for recovery

5) Leg Fracture — Precautions & Recovery

- Immobilize the limb, avoid weight bearing, control bleeding if open fracture
- Pain control + urgent medical evaluation and imaging
- Treatment: casting/splinting vs surgical fixation depending on severity
- Recovery: physical therapy + monitor complications (infection, compartment syndrome)

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Output Evaluation

All five RAG-generated responses were systematically evaluated using LLM-as-judge groundedness and relevance scoring.

Groundedness Evaluation Prompt

Evaluate whether the answer is fully supported by the provided context.
Score from 1–5, where:
1 = Not supported by context
3 = Partially supported
5 = Fully grounded in context.
Provide structured reasoning for the assigned score.

Relevance Evaluation Prompt

Evaluate how directly and completely the answer addresses the user's question.
Score from 1–5, where:
1 = Not relevant
3 = Partially relevant
5 = Fully relevant and complete.
Provide reasoning for the score.

Observations

- RAG improved groundedness compared to LLM-only responses.
- Groundedness scores were sensitive to retrieval coverage (k-value impact).
- Relevance remained consistently high across all queries.
- Partial grounding occurred when retrieved context lacked procedural depth.
- Retrieval quality directly influenced final answer reliability.

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RAG Output Evaluation — Groundedness & Relevance

Scores reflect LLM-as-judge evaluation of RAG answers against retrieved context.

	Query	Groundedness Score (1–5)	Groundedness Verdict	Relevance Score (1–5)	Relevance Verdict
0	Query 1 – Sepsis Management	3	partially_grounded	3	partially_relevant
1	Query 2 – Appendicitis	5	grounded	5	relevant
2	Query 3 – Patchy Hair Loss	4	partially_grounded	3	partially_relevant
3	Query 4 – Traumatic Brain Injury	4	partially_grounded	4	partially_relevant
4	Query 5 – Leg Fracture	5	grounded	5	relevant

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Actionable Insights & Recommendations

RAG Helps Reduce Hallucination Risk

- Grounding responses in trusted medical references reduces hallucination risk
- Improves factual alignment in high-stakes healthcare scenarios
- Enables traceable knowledge sourcing

Retrieval Quality Drives System Performance

- Chunking strategy and k-value directly impact groundedness
- Poor retrieval leads to partial grounding (as seen in some queries)
- Continuous tuning of embeddings and retriever is critical

RAG Enables Domain Specialization Without Retraining

- No model fine-tuning required
- Private or proprietary medical datasets can be integrated securely
- Cost-effective approach compared to full model training

Healthcare Deployment Considerations

- Requires strong evaluation layer (groundedness + relevance monitoring)
- Must include human oversight in clinical settings
- Works best as decision-support, not autonomous diagnosis

Strategic Opportunity

- RAG architecture can be extended to:
- Hospital internal knowledge bases
- Clinical protocols
- Research repositories
- Scalable across healthcare systems and specialty domains

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Data Background and Contents

Data Source

- Merck Medical Manual (PDF)
- ~4,000+ pages of structured medical reference material
- Authoritative clinical resource used by healthcare professionals

Content Coverage

- Symptoms and diagnosis
- Treatment protocols and procedures
- Disease classifications
- Emergency care guidance
- Organ-system-based medical organization (23 sections)

Why This Dataset Was Chosen

- Trusted, medically reviewed source
- Comprehensive coverage across specialties
- Structured content suitable for chunking and retrieval
- Ideal for grounding LLM responses in verified medical knowledge

Data Preparation Summary

- Converted PDF to text
- Split into 1000-character chunks
- 200-character overlap for contextual continuity
- 4,685 chunks indexed in vector database

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